

Exercise Set 2

Exercise 1

a) 2 advantages of multithreading instead of multiprocessing are:

- Lower resource consumption: Threads within the same process share the same memory space and resources
- Faster context switching: Switching between threads is generally quicker than switching between processes due to shared resources.

b) 2 advantages of kernel level threads instead of user level threads are:

- True concurrency: Kernel-level threads can be scheduled independently by the OS, allowing for true parallelism on multi-core processors.
- Better integration with OS services: Kernel threads can take advantage of OS-level features like I/O operations and system calls without the need for additional user-level management.

c) if T is ULT:

- kernel blocks entire process P
- all user-level threads in P are blocked

if T is KLT: - kernel blocks only thread T - other threads in process P can continue executing

Why the difference? - with ULT scheduling is done by user-level library, kernel is unaware of multiple threads within process P. When one thread blocks, the entire process is blocked. With KLT, the kernel is aware of each thread and can manage them independently. The KLT is better in this scenario because it allows other threads in the same process to continue executing while one thread is blocked.

Exercise 2

a)

0-2: A 2-4: B 4-6: C 6-7: D - Finished 7-9: A 9-11: B 11-13: C - Finished 13-14: A - Finished 14-16: B - Finished

Turnaround times: A: 14 B: 16 C: 13 D: 7 Average turnaround time = $(14 + 16 + 13 + 7) / 4 = 12.5$

b)

Turnaround times: A: 5 B: $5 + 6 = 11$ C: $11 + 4 = 15$ D: $15 + 1 = 16$ Average waiting time = $(5 + 11 + 15 + 16) / 4 = 11.75$

c) Order: D, C, A, B Turnaround times: D: 1 C: $1 + 4 = 5$ A: $5 + 5 = 10$ B: $10 + 6 = 16$
Average turnaround time = $(1 + 5 + 10 + 16) / 4 = 8$

Exercise 3

1) SEED 1

Job 0: length = 1, tickets = 84 Job 1: length = 7, tickets = 25 Job 2: length = 4, tickets = 44

Total tickets = 153 Total length = 12

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Random 651593 -> Winning ticket 119 (of 153) -> Run 2
Jobs:
( job:0 timeleft:1 tix:84 ) ( job:1 timeleft:7 tix:25 ) (* job:2 timeleft:4 tix:44 )
Random 788724 -> Winning ticket 9 (of 153) -> Run 0
Jobs:
(* job:0 timeleft:1 tix:84 ) ( job:1 timeleft:7 tix:25 ) ( job:2 timeleft:3 tix:44 )
--> JOB 0 DONE at time 2
Random 93859 -> Winning ticket 19 (of 69) -> Run 1
Jobs:
( job:0 timeleft:0 tix:--- ) (* job:1 timeleft:7 tix:25 ) ( job:2 timeleft:3 tix:44 )
Random 28347 -> Winning ticket 57 (of 69) -> Run 2
Jobs:
( job:0 timeleft:0 tix:--- ) ( job:1 timeleft:6 tix:25 ) (* job:2 timeleft:3 tix:44 )
Random 835765 -> Winning ticket 37 (of 69) -> Run 2
Jobs:
( job:0 timeleft:0 tix:--- ) ( job:1 timeleft:6 tix:25 ) (* job:2 timeleft:2 tix:44 )
Random 432767 -> Winning ticket 68 (of 69) -> Run 2
Jobs:
( job:0 timeleft:0 tix:--- ) ( job:1 timeleft:6 tix:25 ) (* job:2 timeleft:1 tix:44 )
--> JOB 2 DONE at time 6
Random 762280 -> Winning ticket 5 (of 25) -> Run 1
Jobs:
( job:0 timeleft:0 tix:--- ) (* job:1 timeleft:6 tix:25 ) ( job:2 timeleft:0 tix:--- )
Random 2106 -> Winning ticket 6 (of 25) -> Run 1
Jobs:
( job:0 timeleft:0 tix:--- ) (* job:1 timeleft:5 tix:25 ) ( job:2 timeleft:0 tix:--- )
Random 445387 -> Winning ticket 12 (of 25) -> Run 1
Jobs:
( job:0 timeleft:0 tix:--- ) (* job:1 timeleft:4 tix:25 ) ( job:2 timeleft:0 tix:--- )
Random 721540 -> Winning ticket 15 (of 25) -> Run 1
Jobs:
( job:0 timeleft:0 tix:--- ) (* job:1 timeleft:3 tix:25 ) ( job:2 timeleft:0 tix:--- )
Random 228762 -> Winning ticket 12 (of 25) -> Run 1
Jobs:
( job:0 timeleft:0 tix:--- ) (* job:1 timeleft:2 tix:25 ) ( job:2 timeleft:0 tix:--- )
Random 945271 -> Winning ticket 21 (of 25) -> Run 1
Jobs:
( job:0 timeleft:0 tix:--- ) (* job:1 timeleft:1 tix:25 ) ( job:2 timeleft:0 tix:--- )
--> JOB 1 DONE at time 12

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SEED 2

Job 0: length = 9, tickets = 94 Job 1: length = 8, tickets = 73 Job 2: length = 6, tickets = 30

Total tickets = 197 Total length = 23

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Random 605944 -> Winning ticket 169 (of 197) -> Run 2
Jobs:
( job:0 timeleft:9 tix:94 ) ( job:1 timeleft:8 tix:73 ) (* job:2 timeleft:6 tix:30 )
Random 606802 -> Winning ticket 42 (of 197) -> Run 0
Jobs:
(* job:0 timeleft:9 tix:94 ) ( job:1 timeleft:8 tix:73 ) ( job:2 timeleft:5 tix:30 )
Random 581204 -> Winning ticket 54 (of 197) -> Run 0
Jobs:
(* job:0 timeleft:8 tix:94 ) ( job:1 timeleft:8 tix:73 ) ( job:2 timeleft:5 tix:30 )
Random 158383 -> Winning ticket 192 (of 197) -> Run 2
Jobs:
( job:0 timeleft:7 tix:94 ) ( job:1 timeleft:8 tix:73 ) (* job:2 timeleft:5 tix:30 )
Random 430670 -> Winning ticket 28 (of 197) -> Run 0
Jobs:
(* job:0 timeleft:7 tix:94 ) ( job:1 timeleft:8 tix:73 ) ( job:2 timeleft:4 tix:30 )
Random 393532 -> Winning ticket 123 (of 197) -> Run 1
Jobs:
( job:0 timeleft:6 tix:94 ) (* job:1 timeleft:8 tix:73 ) ( job:2 timeleft:4 tix:30 )
Random 723012 -> Winning ticket 22 (of 197) -> Run 0
Jobs:
(* job:0 timeleft:6 tix:94 ) ( job:1 timeleft:7 tix:73 ) ( job:2 timeleft:4 tix:30 )
Random 994820 -> Winning ticket 167 (of 197) -> Run 2
Jobs:
( job:0 timeleft:5 tix:94 ) ( job:1 timeleft:7 tix:73 ) (* job:2 timeleft:4 tix:30 )
Random 949396 -> Winning ticket 53 (of 197) -> Run 0
Jobs:
(* job:0 timeleft:5 tix:94 ) ( job:1 timeleft:7 tix:73 ) ( job:2 timeleft:3 tix:30 )
Random 544177 -> Winning ticket 63 (of 197) -> Run 0
Jobs:
(* job:0 timeleft:4 tix:94 ) ( job:1 timeleft:7 tix:73 ) ( job:2 timeleft:3 tix:30 )
Random 444854 -> Winning ticket 28 (of 197) -> Run 0
Jobs:
(* job:0 timeleft:3 tix:94 ) ( job:1 timeleft:7 tix:73 ) ( job:2 timeleft:3 tix:30 )
Random 268241 -> Winning ticket 124 (of 197) -> Run 1
Jobs:
( job:0 timeleft:2 tix:94 ) (* job:1 timeleft:7 tix:73 ) ( job:2 timeleft:3 tix:30 )
Random 35924 -> Winning ticket 70 (of 197) -> Run 0
Jobs:
(* job:0 timeleft:2 tix:94 ) ( job:1 timeleft:6 tix:73 ) ( job:2 timeleft:3 tix:30 )
Random 27444 -> Winning ticket 61 (of 197) -> Run 0
Jobs:
(* job:0 timeleft:1 tix:94 ) ( job:1 timeleft:6 tix:73 ) ( job:2 timeleft:3 tix:30 )
--> JOB 0 DONE at time 14
Random 464894 -> Winning ticket 55 (of 103) -> Run 1
Jobs:
( job:0 timeleft:0 tix:--- ) (* job:1 timeleft:6 tix:73 ) ( job:2 timeleft:3 tix:30 )
Random 318465 -> Winning ticket 92 (of 103) -> Run 2
Jobs:
( job:0 timeleft:0 tix:--- ) ( job:1 timeleft:5 tix:73 ) (* job:2 timeleft:3 tix:30 )
Random 380015 -> Winning ticket 48 (of 103) -> Run 1
Jobs:
( job:0 timeleft:0 tix:--- ) (* job:1 timeleft:5 tix:73 ) ( job:2 timeleft:2 tix:30 )
Random 891790 -> Winning ticket 16 (of 103) -> Run 1
Jobs:
( job:0 timeleft:0 tix:--- ) (* job:1 timeleft:4 tix:73 ) ( job:2 timeleft:2 tix:30 )
Random 525753 -> Winning ticket 41 (of 103) -> Run 1
Jobs:
( job:0 timeleft:0 tix:--- ) (* job:1 timeleft:3 tix:73 ) ( job:2 timeleft:2 tix:30 )
Random 560510 -> Winning ticket 87 (of 103) -> Run 2
Jobs:
( job:0 timeleft:0 tix:--- ) ( job:1 timeleft:2 tix:73 ) (* job:2 timeleft:2 tix:30 )
Random 236123 -> Winning ticket 47 (of 103) -> Run 1
Jobs:
( job:0 timeleft:0 tix:--- ) (* job:1 timeleft:2 tix:73 ) ( job:2 timeleft:1 tix:30 )
Random 23858 -> Winning ticket 65 (of 103) -> Run 1
Jobs:
( job:0 timeleft:0 tix:--- ) (* job:1 timeleft:1 tix:73 ) ( job:2 timeleft:1 tix:30 )
--> JOB 1 DONE at time 22
Random 325143 -> Winning ticket 3 (of 30) -> Run 2
Jobs:
( job:0 timeleft:0 tix:--- ) ( job:1 timeleft:0 tix:--- ) (* job:2 timeleft:1 tix:30 )
--> JOB 2 DONE at time 23

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SEED 3

Job 0: length = 2, tickets = 54 Job 1: length = 3, tickets = 60 Job 2: length = 6, tickets = 6

Total tickets = 120 Total length = 11

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Random 13168 -> Winning ticket 88 (of 120) -> Run 1
Jobs:
( job:0 timeleft:2 tix:54 ) (* job:1 timeleft:3 tix:60 ) ( job:2 timeleft:6 tix:6 )
Random 837469 -> Winning ticket 109 (of 120) -> Run 1
Jobs:
( job:0 timeleft:2 tix:54 ) (* job:1 timeleft:2 tix:60 ) ( job:2 timeleft:6 tix:6 )
Random 259354 -> Winning ticket 34 (of 120) -> Run 0
Jobs:
(* job:0 timeleft:2 tix:54 ) ( job:1 timeleft:1 tix:60 ) ( job:2 timeleft:6 tix:6 )
Random 234331 -> Winning ticket 91 (of 120) -> Run 1
Jobs:
( job:0 timeleft:1 tix:54 ) (* job:1 timeleft:1 tix:60 ) ( job:2 timeleft:6 tix:6 )
--> JOB 1 DONE at time 4
Random 995645 -> Winning ticket 5 (of 60) -> Run 0
Jobs:
(* job:0 timeleft:1 tix:54 ) ( job:1 timeleft:0 tix:--- ) ( job:2 timeleft:6 tix:6 )
--> JOB 0 DONE at time 5
Random 470263 -> Winning ticket 1 (of 6) -> Run 2
Jobs:
( job:0 timeleft:0 tix:--- ) ( job:1 timeleft:0 tix:--- ) (* job:2 timeleft:6 tix:6 )
Random 836462 -> Winning ticket 2 (of 6) -> Run 2
Jobs:
( job:0 timeleft:0 tix:--- ) ( job:1 timeleft:0 tix:--- ) (* job:2 timeleft:5 tix:6 )
Random 476353 -> Winning ticket 1 (of 6) -> Run 2
Jobs:
( job:0 timeleft:0 tix:--- ) ( job:1 timeleft:0 tix:--- ) (* job:2 timeleft:4 tix:6 )
Random 639068 -> Winning ticket 2 (of 6) -> Run 2
Jobs:
( job:0 timeleft:0 tix:--- ) ( job:1 timeleft:0 tix:--- ) (* job:2 timeleft:3 tix:6 )
Random 150616 -> Winning ticket 4 (of 6) -> Run 2
Jobs:
( job:0 timeleft:0 tix:--- ) ( job:1 timeleft:0 tix:--- ) (* job:2 timeleft:2 tix:6 )
Random 634861 -> Winning ticket 1 (of 6) -> Run 2
Jobs:
( job:0 timeleft:0 tix:--- ) ( job:1 timeleft:0 tix:--- ) (* job:2 timeleft:1 tix:6 )
--> JOB 2 DONE at time 11
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2)

Job 0: length = 10, tickets = 1 Job 1: length = 10, tickets = 100

Total tickets = 101 Total length = 20

The chance that Job 0 gets to run is very low because it has significantly fewer tickets compared to Job 1. It is

$$1 - \left(\frac{100}{101}\right)^{10} \approx 0.095$$

In general with such imbalance jobs will just run in order of the tickets they have.

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Random 844422 -> Winning ticket 62 (of 101) -> Run 1
Jobs:
( job:0 timeleft:10 tix:1 ) (* job:1 timeleft:10 tix:100 )
Random 757955 -> Winning ticket 51 (of 101) -> Run 1
Jobs:
( job:0 timeleft:10 tix:1 ) (* job:1 timeleft:9 tix:100 )
Random 420572 -> Winning ticket 8 (of 101) -> Run 1
Jobs:
( job:0 timeleft:10 tix:1 ) (* job:1 timeleft:8 tix:100 )
Random 258917 -> Winning ticket 54 (of 101) -> Run 1
Jobs:
( job:0 timeleft:10 tix:1 ) (* job:1 timeleft:7 tix:100 )
Random 511275 -> Winning ticket 13 (of 101) -> Run 1
Jobs:
( job:0 timeleft:10 tix:1 ) (* job:1 timeleft:6 tix:100 )
Random 404934 -> Winning ticket 25 (of 101) -> Run 1
Jobs:
( job:0 timeleft:10 tix:1 ) (* job:1 timeleft:5 tix:100 )
Random 783799 -> Winning ticket 39 (of 101) -> Run 1
Jobs:
( job:0 timeleft:10 tix:1 ) (* job:1 timeleft:4 tix:100 )
Random 303313 -> Winning ticket 10 (of 101) -> Run 1
Jobs:
( job:0 timeleft:10 tix:1 ) (* job:1 timeleft:3 tix:100 )
Random 476597 -> Winning ticket 79 (of 101) -> Run 1
Jobs:
( job:0 timeleft:10 tix:1 ) (* job:1 timeleft:2 tix:100 )
Random 583382 -> Winning ticket 6 (of 101) -> Run 1
Jobs:
( job:0 timeleft:10 tix:1 ) (* job:1 timeleft:1 tix:100 )
--> JOB 1 DONE at time 10
Random 908113 -> Winning ticket 0 (of 1) -> Run 0
Jobs:
(* job:0 timeleft:10 tix:1 ) ( job:1 timeleft:0 tix:--- )
Random 504687 -> Winning ticket 0 (of 1) -> Run 0
Jobs:
(* job:0 timeleft:9 tix:1 ) ( job:1 timeleft:0 tix:--- )
Random 281838 -> Winning ticket 0 (of 1) -> Run 0
Jobs:
(* job:0 timeleft:8 tix:1 ) ( job:1 timeleft:0 tix:--- )
Random 755804 -> Winning ticket 0 (of 1) -> Run 0
Jobs:
(* job:0 timeleft:7 tix:1 ) ( job:1 timeleft:0 tix:--- )
Random 618369 -> Winning ticket 0 (of 1) -> Run 0
Jobs:
(* job:0 timeleft:6 tix:1 ) ( job:1 timeleft:0 tix:--- )
Random 250506 -> Winning ticket 0 (of 1) -> Run 0
Jobs:
(* job:0 timeleft:5 tix:1 ) ( job:1 timeleft:0 tix:--- )
Random 909747 -> Winning ticket 0 (of 1) -> Run 0
Jobs:
(* job:0 timeleft:4 tix:1 ) ( job:1 timeleft:0 tix:--- )
Random 982786 -> Winning ticket 0 (of 1) -> Run 0
Jobs:
(* job:0 timeleft:3 tix:1 ) ( job:1 timeleft:0 tix:--- )
Random 810218 -> Winning ticket 0 (of 1) -> Run 0
Jobs:
(* job:0 timeleft:2 tix:1 ) ( job:1 timeleft:0 tix:--- )
Random 902166 -> Winning ticket 0 (of 1) -> Run 0
Jobs:
(* job:0 timeleft:1 tix:1 ) ( job:1 timeleft:0 tix:--- )
--> JOB 0 DONE at time 20

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Exercise 4

- 1) 30 ms
- 2) 20 ms
- 3) Second column decrease twice as fast after warmup period.
- 4) After 9

Exercise 5

- 5) 150 ms scheduler destroys cache locality
- 6) 110 ms Runtime significantly improves
- 7)

Cache size = 50 With 1 CPU • All jobs always cold • Very slow

With 2 CPUs • Still cold, but some parallelism

With 3 CPUs • Still cold, but more parallelism

Speedup is sub-linear or roughly linear.

Cache size = 100 With 1 CPU • Jobs evict each other • Mostly cold

With 2 CPUs • Some improvement • Still interference

With 3 CPUs • Each job gets its own CPU • Each cache warms • All jobs run fast